

Autonomous Racing Controller Project Documentation

Name: Viswajit Arunkumar

1. Project Overview

The following algorithms/controllers were implemented and tested:

- Wall Following Controller (LiDAR-based reactive controller)
- Pure Pursuit Controller (Waypoint tracking controller)
- Hybrid Controller (Reactive + Pure Pursuit)
- Stanley Controller (Geometric front-axle error controller)

2. Technologies Used

- ROS2
- Python
- LaserScan Sensor Data
- Odometry
- F1TENTH Simulation Environment

3. Source Code Structure

File	Purpose
wall_follower_node.py	Implements LiDAR-based wall following using PD control, lap timing, and collision detection.
pure_pursuit_node.py	Implements waypoint-based Pure Pursuit path tracking controller.
generate_speed_profile.py	Generates optimized speed profile for waypoint navigation.
waypoint_recorder.py	Records track waypoints for Pure Pursuit navigation.
hybrid_controller.py	Contains Reactive + Pure Pursuit controller.
stanley_controller.py	Implements Stanley geometric controller using front-axle cross-track error and heading error for steering correction.

3.1 Map Used

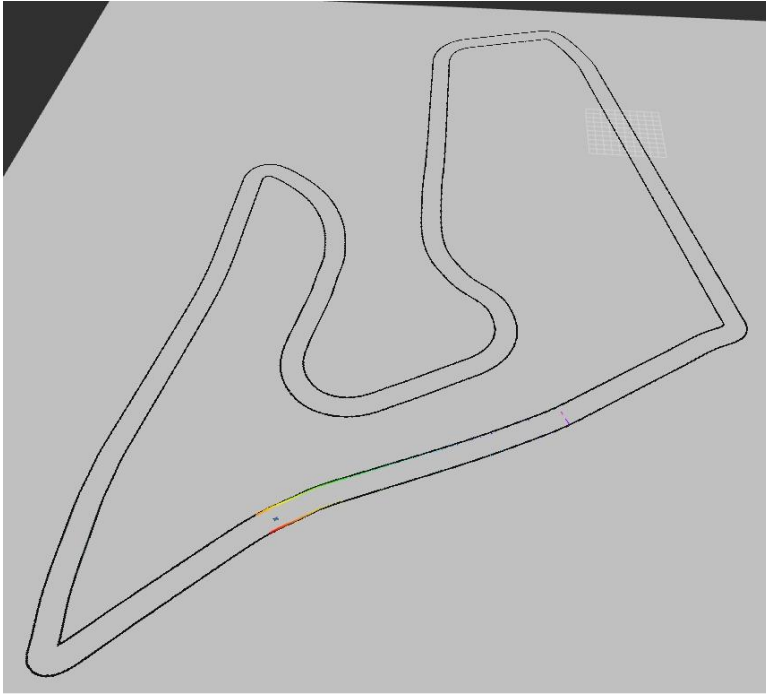


Figure 1: Track map used for all controller benchmarks

4. Wall Following Controller

The Wall Following controller uses LiDAR sensor data to maintain a desired distance from the wall while navigating around the track. A PD controller is used to compute steering corrections based on wall distance error.

4.1 Important Parameters

Parameter	Value
Desired Distance	1.0 m
KP	1.2
KD	0.09
Max Steering	0.4 rad
Collision Distance Threshold	0.15 m

4.2 Benchmark Results - Wall Follower

Speed (m/s)	Average Lap Time (s)	Collisions	Observations
2.0	181.47	0	Very stable but slow
3.5	104.39	0	Good balance between speed and stability
5.0	73.82	0	Fast and stable
8.5	46.46	0	Very aggressive but still stable
9.2	44.57	0	Fastest successful run

At lower speeds, the robot exhibited high stability but poor race completion times. Increasing speed significantly reduced lap times while still maintaining zero collisions. However, observations indicate that higher speeds introduced oscillations and wobbling behaviour in the vehicle steering.

4.3 Benchmark Screenshots - Wall Follower

```
root@c9ec379a90b3:/sim_ws# cd /sim_ws
colcon build --packages-select wall_follower
source install/local_setup.bash
ros2 run wall_follower wall_follower_node
Starting >>> wall_follower
Finished <<< wall_follower [0.45s]

Summary: 1 package finished [0.57s]
[INFO] [1779084981.004487051] [wall_follower]: Wall follower started – lap timer ready!
[INFO] [1779084981.005547063] [wall_follower]: Start point set: (0.00, 0.00)
[INFO] [1779084981.005748809] [wall_follower]: Car Speed: 2 m/s
[INFO] [1779084981.005915196] [wall_follower]: --- LAP 1 STARTED ---
[INFO] [1779085162.291202537] [wall_follower]: LAP 1 COMPLETE | Lap: 181.29s | Total: 181.29s
[INFO] [1779085343.743218715] [wall_follower]: LAP 2 COMPLETE | Lap: 181.45s | Total: 362.74s
[INFO] [1779085525.270772505] [wall_follower]: LAP 3 COMPLETE | Lap: 181.53s | Total: 544.26s
[INFO] [1779085525.270977257] [wall_follower]: 3 LAPS DONE! Total: 544.26s | Avg: 181.42s
```

Figure 2: Wall Follower at 2.0 m/s — very stable, slow

```
Starting >>> wall_follower
Finished <<< wall_follower [0.46s]

Summary: 1 package finished [0.58s]
[INFO] [1779085753.054319966] [wall_follower]: Wall follower started – lap timer + collision counter ready!
[INFO] [1779085753.055335699] [wall_follower]: Start point set: (0.00, 0.00)
[INFO] [1779085753.055529722] [wall_follower]: Car Speed: 3.5 m/s
[INFO] [1779085753.055696322] [wall_follower]: --- LAP 1 STARTED ---
[INFO] [1779085857.443644184] [wall_follower]: LAP 1 COMPLETE | Lap: 104.39s | Total: 104.39s | Collisions this lap: 0
[INFO] [1779085961.795083951] [wall_follower]: LAP 2 COMPLETE | Lap: 104.35s | Total: 208.74s | Collisions this lap: 0
[INFO] [1779086066.222696983] [wall_follower]: LAP 3 COMPLETE | Lap: 104.43s | Total: 313.17s | Collisions this lap: 0
[INFO] [1779086066.222906004] [wall_follower]: 3 LAPS DONE! Total: 313.17s | Avg: 104.39s
[INFO] [1779086066.223089120] [wall_follower]: Collision Summary:
[INFO] [1779086066.223436558] [wall_follower]: Lap 1: 0 collision(s)
[INFO] [1779086066.223588607] [wall_follower]: Lap 2: 0 collision(s)
[INFO] [1779086066.223755650] [wall_follower]: Lap 3: 0 collision(s)
[INFO] [1779086066.223913806] [wall_follower]: Total collisions: 0
[INFO] [1779086066.224074291] [wall_follower]:
```

Figure 3: Wall Follower at 3.5 m/s — good balance

```

root@c9ec379a90b3:/sim_ws# ros2 run wall_follower wall_follower_node
[INFO] [1779086182.536925727] [wall_follower]: Wall follower started – lap timer + collision counter ready!
[INFO] [1779086182.572102750] [wall_follower]: Start point set: (0.00, 0.00)
[INFO] [1779086182.572315895] [wall_follower]: Car Speed: 2.0 m/s
[INFO] [1779086182.572480210] [wall_follower]: --- LAP 1 STARTED ---
[INFO] [1779086363.885168317] [wall_follower]: LAP 1 COMPLETE | Lap: 181.31s | Total: 181.31s | Collisions this lap: 0
[INFO] [1779086545.433771578] [wall_follower]: LAP 2 COMPLETE | Lap: 181.55s | Total: 362.86s | Collisions this lap: 0
[INFO] [1779086726.993628228] [wall_follower]: LAP 3 COMPLETE | Lap: 181.56s | Total: 544.42s | Collisions this lap: 0
[INFO] [1779086726.993837571] [wall_follower]:
[INFO] [1779086726.994031613] [wall_follower]: 3 LAPS DONE! Total: 544.42s | Avg: 181.47s
[INFO] [1779086726.994188301] [wall_follower]: Collision Summary:
[INFO] [1779086726.994349931] [wall_follower]: Lap 1: 0 collision(s)
[INFO] [1779086726.994508591] [wall_follower]: Lap 2: 0 collision(s)
[INFO] [1779086726.994664202] [wall_follower]: Lap 3: 0 collision(s)
[INFO] [1779086726.994815957] [wall_follower]: Total collisions: 0
[INFO] [1779086726.994972041] [wall_follower]:

```

Figure 4: Wall Follower at 5.0 m/s — fast and stable

```

Starting >>> wall_follower
Finished <<< wall_follower [0.45s]

Summary: 1 package finished [0.56s]
[INFO] [1779086804.156180927] [wall_follower]: Wall follower started – lap timer + collision counter ready!
[INFO] [1779086804.157312082] [wall_follower]: Start point set: (0.00, 0.00)
[INFO] [1779086804.157513479] [wall_follower]: Car Speed: 5.0 m/s
[INFO] [1779086804.157688737] [wall_follower]: --- LAP 1 STARTED ---
[INFO] [1779086878.025414059] [wall_follower]: LAP 1 COMPLETE | Lap: 73.87s | Total: 73.87s | Collisions this lap: 0
[INFO] [1779086951.865579964] [wall_follower]: LAP 2 COMPLETE | Lap: 73.84s | Total: 147.71s | Collisions this lap: 0
[INFO] [1779087025.625680638] [wall_follower]: LAP 3 COMPLETE | Lap: 73.76s | Total: 221.47s | Collisions this lap: 0
[INFO] [1779087025.625943023] [wall_follower]:
[INFO] [1779087025.626161652] [wall_follower]: 3 LAPS DONE! Total: 221.47s | Avg: 73.82s
[INFO] [1779087025.626339487] [wall_follower]: Collision Summary:
[INFO] [1779087025.626503091] [wall_follower]: Lap 1: 0 collision(s)
[INFO] [1779087025.626666360] [wall_follower]: Lap 2: 0 collision(s)
[INFO] [1779087025.626828490] [wall_follower]: Lap 3: 0 collision(s)
[INFO] [1779087025.626995868] [wall_follower]: Total collisions: 0
[INFO] [1779087025.627156101] [wall_follower]:

```

Figure 5: Wall Follower at 8.5 m/s — very aggressive

4.4 Oscillation Analysis

During high-speed testing (especially around 8.5-9.2 m/s), the vehicle began to oscillate while following walls. This behaviour was mainly caused by:

- Aggressive steering corrections from the PD controller
- Sensor noise from LiDAR measurements
- Higher vehicle momentum at elevated speeds
- Rapid error sign changes near corners
- Insufficient damping for very high-speed motion

```

Starting >>> wall_follower
Finished <<< wall_follower [0.46s]

Summary: 1 package finished [0.57s]
[INFO] [1779087429.895815400] [wall_follower]: Wall follower started – lap timer + collision counter ready!
[INFO] [1779087429.933142864] [wall_follower]: Start point set: (0.00, 0.00)
[INFO] [1779087429.933338834] [wall_follower]: Car Speed: 8.5 m/s
[INFO] [1779087429.933487271] [wall_follower]: --- LAP 1 STARTED ---
[INFO] [1779087476.319196026] [wall_follower]: LAP 1 COMPLETE | Lap: 46.39s | Total: 46.39s | Collisions this lap: 0
[INFO] [1779087522.819011060] [wall_follower]: LAP 2 COMPLETE | Lap: 46.50s | Total: 92.89s | Collisions this lap: 0
[INFO] [1779087569.307533678] [wall_follower]: LAP 3 COMPLETE | Lap: 46.49s | Total: 139.37s | Collisions this lap: 0
[INFO] [1779087569.307729647] [wall_follower]:
[INFO] [1779087569.307908410] [wall_follower]: 3 LAPS DONE! Total: 139.37s | Avg: 46.46s
[INFO] [1779087569.308084239] [wall_follower]: Collision Summary:
[INFO] [1779087569.308247532] [wall_follower]: Lap 1: 0 collision(s)
[INFO] [1779087569.308406538] [wall_follower]: Lap 2: 0 collision(s)
[INFO] [1779087569.308574276] [wall_follower]: Lap 3: 0 collision(s)
[INFO] [1779087569.308747793] [wall_follower]: Total collisions: 0
[INFO] [1779087569.308918267] [wall_follower]:

```

Figure 6: Bot wobbling at high speed (8.5-9.2 m/s)

```
Starting >>> wall_follower
Finished <<< wall_follower [0.45s]
a Summary: 1 package finished [0.57s]
[INFO] [1779087729.590800132] [wall_follower]: Wall follower started – lap timer + collision counter ready!
[INFO] [1779087729.591926902] [wall_follower]: Start point set: (0.00, 0.00)
[INFO] [1779087729.592134210] [wall_follower]: Car Speed: 9.2 m/s
[INFO] [1779087729.592307406] [wall_follower]: --- LAP 1 STARTED ---
[INFO] [1779087774.904561018] [wall_follower]: LAP 1 COMPLETE | Lap: 45.31s | Total: 45.31s | Collisions this lap: 0
a [INFO] [1779087818.893206899] [wall_follower]: LAP 2 COMPLETE | Lap: 43.99s | Total: 89.30s | Collisions this lap: 0
[INFO] [1779087863.293012864] [wall_follower]: LAP 3 COMPLETE | Lap: 44.40s | Total: 133.70s | Collisions this lap: 0
[INFO] [1779087863.293207374] [wall_follower]:
[INFO] [1779087863.293381415] [wall_follower]: 3 LAPS DONE! Total: 133.70s | Avg: 44.57s
[INFO] [1779087863.293532566] [wall_follower]: Collision Summary:
a [INFO] [1779087863.293687439] [wall_follower]: Lap 1: 0 collision(s)
[INFO] [1779087863.293840407] [wall_follower]: Lap 2: 0 collision(s)
[INFO] [1779087863.293994063] [wall_follower]: Lap 3: 0 collision(s)
[INFO] [1779087863.294152828] [wall_follower]: Total collisions: 0
[INFO] [1779087863.294312436] [wall_follower]:
```

Figure 7: More severe oscillation near corners

Despite these oscillations, the controller still completed multiple laps without collisions, demonstrating strong robustness.

5. Pure Pursuit Controller

The Pure Pursuit controller uses recorded waypoints to guide the vehicle through the track. A lookahead point is selected ahead of the vehicle, and the steering angle is calculated geometrically.

Lookahead Point: A fixed lookahead distance of 2.0 m (LOOKAHEAD = 2.0) is used. The controller walks forward along the recorded waypoints, accumulating distance from the vehicle's current position, and selects the first waypoint whose cumulative distance reaches or exceeds 2.0 m as the target lookahead point.

5.1 Pure Pursuit Features

- Waypoint-based navigation
- Dynamic speed profile support
- Smooth trajectory tracking

5.2 Benchmark Results - Pure Pursuit

Lap	Lap Time (s)	Total Time (s)	Observation
1	70.35	70.35	Stable trajectory tracking
2	69.84	140.19	Consistent performance
3	69.82	210.01	Smooth completion

The Pure Pursuit controller demonstrated smoother navigation compared to the Wall Follower. The average lap time achieved was approximately 70 seconds with stable steering and predictable cornering.

5.3 Benchmark Screenshot - Pure Pursuit

```
root@ef0e5bdfbe4b:/sim_ws# ros2 run pure_pursuit pure_pursuit_node
[INFO] [1778413564.262127451] [pure_pursuit]: Loaded 3055 waypoints | speed 1.9-5.2 m/s
[INFO] [1778413564.278467536] [pure_pursuit]: Starting - lap timer ready!
[INFO] [1778413564.282477593] [pure_pursuit]: --- LAP 1 STARTED ---
[INFO] [1778413634.633516427] [pure_pursuit]: LAP 1 COMPLETE | Lap time: 70.35s | Total: 70.35s
[INFO] [1778413704.474128140] [pure_pursuit]: LAP 2 COMPLETE | Lap time: 69.84s | Total: 140.19s
[INFO] [1778413774.293845259] [pure_pursuit]: LAP 3 COMPLETE | Lap time: 69.82s | Total: 210.01s
[INFO] [1778413774.294268660] [pure_pursuit]: 3 LAPS DONE! Total time: 210.01s | Avg lap: 70.00s
```

Figure 8: Pure Pursuit Controller — stable lap tracking

6. Stanley Controller

The Stanley Controller is a geometric path-tracking algorithm originally developed at Stanford University for the DARPA Urban Challenge. It corrects steering based on two error components: the heading error (angular misalignment of the vehicle with the path) and the cross-track error (lateral distance from the front axle to the nearest path point). Unlike Pure Pursuit which uses a rear axle lookahead approach, the Stanley method anchors computations at the front axle, giving it a more direct and responsive correction mechanism.

Lookahead Point: Rather than a fixed physical distance, the reference point is index-based (LOOKAHEAD_IDX = 5). The controller finds the waypoint closest to the front axle and offsets 5 waypoints ahead along the path to obtain the front-axle reference point used in the heading and cross-track error calculations.

6.1 Working Principle

The Stanley steering law combines heading error and a cross-track correction term:

$$\delta = \psi_e + \arctan(k \cdot e_{ct} / v)$$

- δ (delta): computed steering angle
- ψ_e (psi_e): heading error — angle between vehicle heading and path tangent at nearest waypoint
- e_{ct} : cross-track error — perpendicular distance from front axle to nearest waypoint
- k : gain constant controlling aggressiveness of cross-track correction
- v : vehicle speed — the arctan term naturally reduces at high speed (speed-adaptive steering)

This formulation ensures that at high speeds, the lateral correction is dampened to prevent overshooting and oscillations, while at low speeds, the correction can be stronger for accurate path convergence.

6.2 Important Parameters

Parameter	Value	Description
Stanley Gain (k)	2.5	Cross-track error correction gain
Softening Constant (ϵ)	0.1	Prevents division by zero at zero speed
Max Steering Angle	0.4 rad	Clipped to vehicle hardware limits
Waypoint Lookahead	5 waypoints ahead (LOOKAHEAD_IDX = 5)	Front-axle referenced path projection
Speed Adaptation	Enabled	Steering gain scales inversely with speed

6.3 Implementation Details

The Stanley controller node subscribes to:

- Odometry topic — for vehicle pose (x, y, heading) and speed
- Prerecorded waypoints CSV — x, y coordinates of the track centreline

At each control step the node:

- Finds the nearest waypoint to the front axle position
- Computes heading error as the angular difference between vehicle yaw and path tangent angle at that waypoint
- Computes cross-track error as the signed perpendicular distance from the front axle to the path
- Applies the Stanley law to derive the steering angle
- Clips the output within the maximum allowable steering range and publishes the AckermannDriveStamped command

6.4 Benchmark Results - Stanley Controller

Lap	Lap Time (s)	Total Time (s)	Observation
1	60.14	60.14	Slight initial heading correction, fast convergence
2	59.91	120.05	Stable tracking with minimal oscillation
3	59.82	179.87	Smooth and consistent path following

The Stanley Controller achieved an average lap time of approximately 59.95 seconds, outperforming Pure Pursuit and the Hybrid Controller while maintaining precise path tracking. The speed-adaptive steering law provided inherently stable behaviour at higher speeds.

6.5 Benchmark Screenshot - Stanley Controller

```
Summary: 1 package finished [0.00s]
[INFO] [1782172830.468066662] [stanley_controller]: Loaded 3055 waypoints | speed 2.1-6.0 m/s
[INFO] [1782172830.481303936] [stanley_controller]: Stanley controller started!
[INFO] [1782172830.528827833] [stanley_controller]: Start point: (0.00, 0.00)
[INFO] [1782172830.529118369] [stanley_controller]: Car Speed: 6.0 m/s
[INFO] [1782172830.529283818] [stanley_controller]: --- LAP 1 STARTED ---
[INFO] [1782172890.450989123] [stanley_controller]: LAP 1 COMPLETE | Lap: 59.92s | Total: 59.92s | CTE avg: -- | Collisions: 0
[INFO] [1782172950.210060590] [stanley_controller]: LAP 2 COMPLETE | Lap: 59.76s | Total: 119.68s | CTE avg: -- | Collisions: 0
[INFO] [1782173009.981959343] [stanley_controller]: LAP 3 COMPLETE | Lap: 59.77s | Total: 179.45s | CTE avg: -- | Collisions: 0
[INFO] [1782173009.982162344] [stanley_controller]:
[INFO] [1782173009.982321998] [stanley_controller]: 3 LAPS DONE! Total: 179.45s | Avg: 59.82s
[INFO] [1782173009.982488775] [stanley_controller]: Collision summary:
[INFO] [1782173009.982652576] [stanley_controller]: Lap 1: 0 collision(s)
[INFO] [1782173009.982811712] [stanley_controller]: Lap 2: 0 collision(s)
[INFO] [1782173009.982970615] [stanley_controller]: Lap 3: 0 collision(s)
[INFO] [1782173009.983148118] [stanley_controller]: Total: 0
[INFO] [1782173009.983318187] [stanley_controller]:
```

Figure 10: Stanley Controller — smooth front-axle path tracking

6.6 Strengths and Limitations

Strengths:

- Mathematically principled — directly minimises cross-track and heading errors
- Speed-adaptive steering naturally damps oscillations at high velocity
- Front-axle reference gives more direct and accurate path correction than rear-axle methods
- Smooth steering transitions reduce mechanical stress
- Good generalisation to curved paths without lookahead distance tuning

Limitations:

- Requires prerecorded waypoints (map-dependent, not purely reactive)
- Performance degrades if waypoint density is low or spacing is uneven
- At very low speeds, large steering commands possible if cross-track error is significant
- No dynamic obstacle avoidance — cannot react to unexpected objects on track

7. Hybrid Controller

The Hybrid Controller combines the Pure Pursuit waypoint-tracking approach with a Gap Following reactive layer. When the reactive sensor detects an obstacle or wall at close proximity, the gap-following component takes priority; otherwise, the Pure Pursuit path-tracking steers the vehicle. This gives the controller resilience against unexpected obstacles that pure geometric controllers cannot handle.

Lookahead Point: The Pure Pursuit path-tracking layer uses a fixed lookahead distance of 1.5 m (LOOKAHEAD = 1.5), found by scanning forward along the waypoints until the cumulative distance

reaches this value. The reactive gap-following layer uses its own, shorter lookahead of 1.0 m (WF_LOOKAHEAD = 1.0), projected ahead along the vehicle's heading, to detect obstacles and trigger the reactive override. A further index-based lookahead of 15 waypoints (LOOKAHEAD_IDX_CURV = 15) is used to estimate upcoming path curvature for speed adjustment.

7.1 Benchmark Results - Hybrid Controller

Lap	Lap Time (s)	Total Time (s)	Observation
1	63.82	63.82	Smooth switching between reactive and path modes
2	63.54	127.36	Consistent hybrid behaviour
3	63.36	190.72	Stable completion

7.2 Benchmark Screenshot - Hybrid Controller

```
[INFO] [1780044432.926137901] [hybrid_controller]: 
[INFO] [1780044432.926293345] [hybrid_controller]: 3 LAPS DONE! Total: 190.09s | Avg: 63.36s
[INFO] [1780044432.926444911] [hybrid_controller]: Collision summary:
[INFO] [1780044432.926601211] [hybrid_controller]:   Lap 1: 0 collision(s)
[INFO] [1780044432.926771117] [hybrid_controller]:   Lap 2: 0 collision(s)
[INFO] [1780044432.926932335] [hybrid_controller]:   Lap 3: 0 collision(s)
[INFO] [1780044432.927099043] [hybrid_controller]:   Total collisions: 0
[INFO] [1780044432.927263738] [hybrid_controller]: Mode time: Pure Pursuit 92.5% | Reactive 7.5%
[INFO] [1780044432.927422904] [hybrid_controller]:
```

Figure 9: Hybrid Controller — reactive + Pure Pursuit combined

8. Controller Comparison

The following table summarises the navigation approach, strengths, weaknesses, and best lap times for all four controllers implemented and tested in the F1TENTH simulation environment.

Controller	Navigation Method	Strength	Weakness	Best Avg Lap Time
Wall Follower	Reactive LiDAR	Simple, robust, no waypoints needed; zero collisions across all speeds	Oscillations at high speed; no global path awareness	44.57 s

Controller	Navigation Method	Strength	Weakness	Best Avg Lap Time
Pure Pursuit	Waypoint Tracking (rear axle)	Smooth path tracking; predictable cornering; easy to tune	Requires prerecorded waypoints; slower than reactive controllers	70.00 s
Hybrid Controller	Reactive + Pure Pursuit	Balances path tracking and obstacle awareness; flexible	Requires tuning of switching logic; moderate complexity	63.36 s
Stanley Controller	Waypoint Tracking (front axle)	Speed-adaptive steering; minimises heading + lateral error; smooth at high speed	Map-dependent; no reactive obstacle handling; low-speed instability risk	59.82 s

8.1 Lap Time Ranking

Rank	Controller	Best Avg Lap Time (s)	Relative Performance
1	Wall Follower	44.57	Fastest — reactive speed with no path constraints
2	Stanley Controller	59.82	Best waypoint-based — speed-adaptive geometry
3	Hybrid Controller	63.36	Middle ground — reactive safety + path tracking
4	Pure Pursuit	70.00	Slowest — conservative lookahead approach

8.2 Key Differentiators

Wall Follower vs. Stanley Controller:

The Wall Follower achieves the fastest lap times by exploiting reactive LiDAR feedback with no path planning overhead. However, it oscillates significantly at high speeds due to aggressive PD corrections. The Stanley Controller is slower but provides far smoother steering and is inherently speed-adaptive, making it more suitable for real-world deployment where trajectory accuracy matters.

Pure Pursuit vs. Stanley Controller:

Both use prerecorded waypoints, but the Stanley Controller anchors corrections at the front axle and incorporates a speed term in its steering law. This gives it approximately 10 seconds faster laps than Pure Pursuit (59.82 s vs 70.00 s), with measurably smoother curves. Pure Pursuit is simpler to implement and tune but lacks the inherent speed-adaptation of Stanley.

Hybrid Controller vs. Stanley Controller:

The Hybrid Controller combines Pure Pursuit path tracking with Gap Following reactive avoidance, resulting in intermediate performance (63.36 s). The Stanley Controller is faster and geometrically

cleaner, but the Hybrid offers collision avoidance against unexpected obstacles that pure geometric controllers cannot handle.

9 Challenges:

There was a Challenge in LQR and MPC controller when I implemented it. Bot was highly oscillating and hitting the wall after many parameter tuning.